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Coupled interfacial desolvation regulation and LiF-Rich SEI for fast-charging lithium metal batteries in carbonate electrolytes

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ABSTRACT

Lithium metal anode is considered as the most promising candidates for next-generation batteries due to its high specific energy. However, fast charging of lithium metal batteries in widely used carbonate electrolytes remains highly challenging, as insufficient interfacial desolvation and non-ideal solid electrolyte interphase (SEI) formation compromise lithium deposition uniformity. Here, we demonstrate that fast-charging lithium metal batteries can be enabled through a coupled regulation of interfacial desolvation behavior and LiF-rich SEI formation, achieved by introducing a stable molecular layer at the lithium surface without altering the bulk electrolyte composition for high Li⁺ conductivity. Consequently, during the fast-charging process, lithium metal maintains the spherical morphology of deposition, even at the ultra-high current density of 6 mA cm⁻². And the dense dendrite-free lithium growth can still retain under the ultra-high deposition capacity of 100 mAh cm⁻². The assembled Li||LFP full cell delivers high capacity retention of 85% after 3000 cycles under ultra-high rate of 20 C. This work offers insights on unlocking the fast-charging potential of lithium metal batteries in high-rate applications.

1. Introduction

With the growing demand for fast-charging, long-range electric vehicles (EVs), the development of batteries with both high energy density and high power capability has become increasingly urgent [1–4]. Lithium metal anodes (LMAs), featuring an ultrahigh theoretical specific capacity (3860 mAh g⁻¹) and a low redox potential (−3.04 V, vs.SHE), are widely regarded as promising candidates for next-generation high-energy batteries. Beyond energy density, fast charging at rates comparable to conventional refueling is essential for alleviating EV range anxiety [5–7], and improving user acceptance [8,9].

However, the slow Li⁺ transport during the rapid charging process of LMAs leads to severe lithium dendrite growth, deteriorating the cycle life of batteries [10,11]. The Li⁺ transport is divided into two parts, the electrolyte phase transport and the electrode/electrolyte interface

transport. Strongly solvating carbonate solvents are commonly used to ensure efficient Li⁺ transport in the bulk electrolyte by promoting lithium salt dissociation and mobility [12,13]. However, excessive solvation makes Li⁺ desolvation at the electrode surface more difficult, thereby hindering interfacial charge transfer. Although extensive efforts have been devoted to regulating SEI composition for fast ion transport [14–17], the impact of strong electrolyte solvation on interfacial Li⁺ desolvation has not been adequately addressed.

Fluorosilane compounds have attracted attention for lithium metal interface modification owing to their strong affinity to lithium surfaces and their ability to form robust interfacial layers with high ionic conductivity and mechanical stability [18,19]. And Cui et al. reported polysiloxane branches to form a salt-philic, solvent-phobic interface layer for anion-derived SEI [20]. While the fast-charging performance of these silane-modified Li metals has not been fully demonstrated, these

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studies highlight the potential of fluorosilane-based molecular design for regulating interfacial solvation environments and SEI chemistry, which may be particularly advantageous for addressing the trade-offs among bulk electrolyte transport, interfacial desolvation, and SEI properties during fast charging.

Herein, a synergistic desolvation and LiF-rich SEI chemistry is achieved by stable molecular layer on the surface of LMAs. This synergistic regulation modulates the interfacial solvation environment, inducing repulsive interactions toward solvent molecules within the Li^+ solvation sheath and thereby facilitating interfacial desolvation (Fig. 1a). Simultaneously, the modified solvation environment promotes increased anion participation near the lithium surface, favoring the formation of a compact, LiF-rich SEI that supports efficient ion transport across the

interphase. Importantly, this interfacial regulation is achieved in strongly solvating carbonate electrolytes, thus preserving high ionic conductivity in the electrolyte phase. As a result, a continuous and abundant Li^+ supply near the lithium anode is achieved for matching the rapid lithium deposition reactions, resulting in spherical lithium deposition. And the spherical morphology can be maintained even at the ultra-high current density of 6 mA cm^{-2} . The dense dendrite-free lithium growth is also achieved at the ultra-high deposition surface capacity of 100 mAh cm^{-2} . The $\text{Li}||\text{LiFePO}_4(\text{LFP})$ full cell exhibits 85% capacity retention after 3000 cycles at ultra-high rate of 20 C.

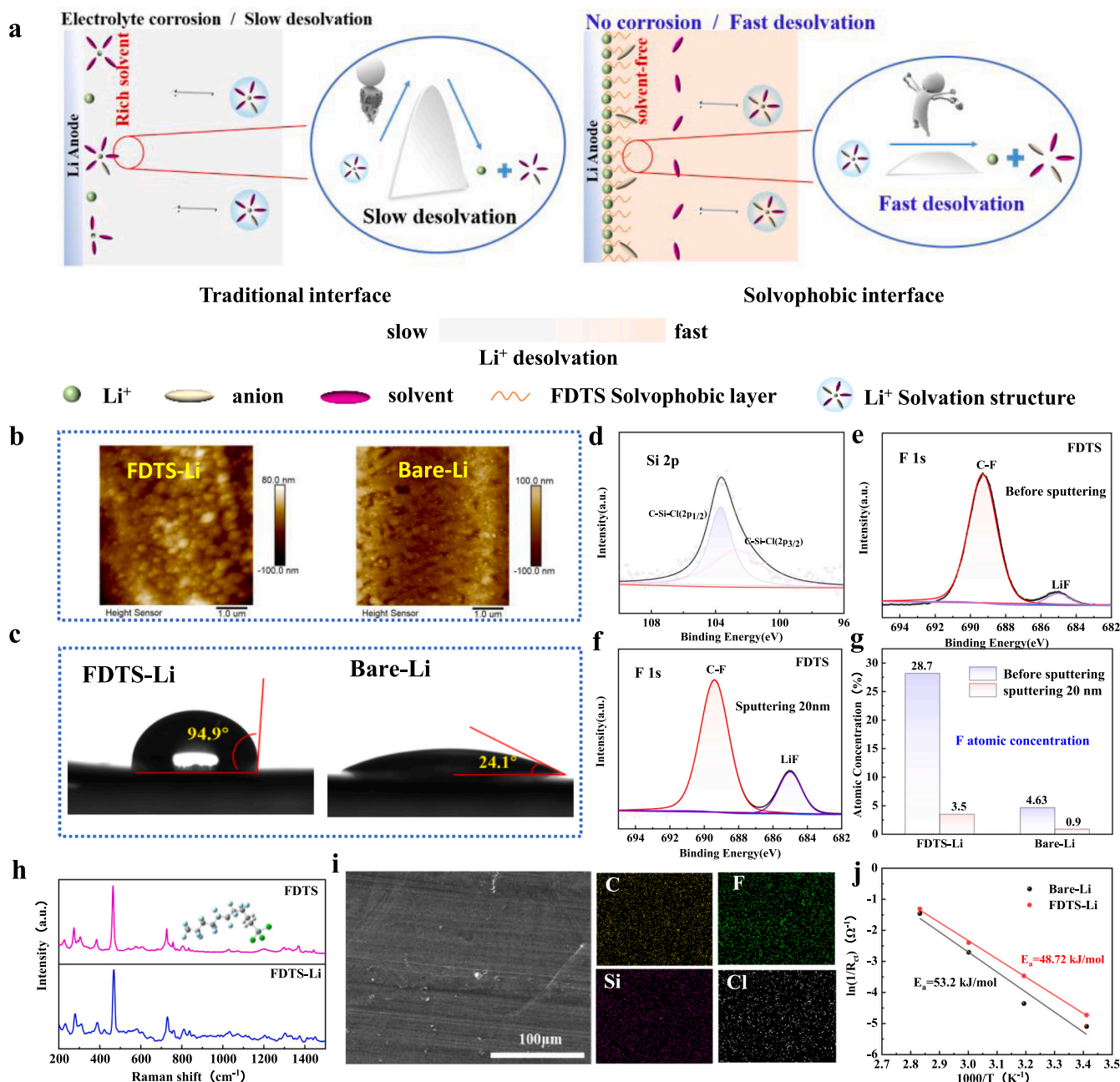


Fig. 1. Preparation and structure evaluation of FDTD-Li anodes. **a** Schematic diagram of the desolvation process at the control interface of the solvophobic layer. **b** AFM tests for FDTD-Li and Bare-Li. **c** Contact angle test of electrolyte on FDTD-Li and Bare-Li. **d** Si 2p and **e** F 1s XPS spectra of FDTD-Li after 50 cycles. **f** F 1s XPS spectra of FDTD-Li after 50 cycles. The surface of the sample was sputtered by 20 nm. **g** Surface F atom content of FDTD-Li and Bare-Li before and after 20 nm sputtering. **h** Raman spectrum of FDTD and FDTD-Li. **i** SEM image of the FDTD-Li anode and distributions of C, F, Si, and Cl elements obtained from EDS tests. **j** Desolvation energy (E_d) of Bare-Li and FDTD-Li.

2. Results and discussion

This work uses the solvophobic 1H, 1H, 2H, 2H- perfluorodecyltri-chlorosilane (FDTS) as a lithium metal anode protective layer, and the solvophobic effect compared with previous work has significant performance advantages (Figure S1) [18–22]. The formation of a solvophobic FDTS coating (Figure S2) on lithium metal is achieved by applying FDTS solution to the metal surface and allowing it to evaporate slowly in a glove box (Figure S3). The lithium metal coated with FDTS, referred to as FDTS-Li. Atomic force microscopy (AFM) analysis in Fig. 1b corroborates the uniform distribution of the FDTS. To examine the solvophobic character of FDTS coating toward electrolyte, contact angle tests were conducted on both fresh and treated Li, and Cu substrates. The electrolyte used in this work consists of 1 M LiPF₆ in EC/DEC (1:1, w/w) with 10% FEC. As shown in Fig. 1c and Figure S4. The contact angle between the FDTS-Li surface and the electrolyte is 94.9°, higher than 24.1° for Bare-Li. Similarly, the contact angle between FDTS-Cu and the electrolyte is 64.6°, while the Bare-Cu exhibits a contact angle of 0° in a fully wetted state. Excessive wettability of the electrolyte with lithium metal can lead to increased side reactions between the solvents and lithium, resulting in higher electrolyte consumption and the formation of dead lithium [23–26]. Hence, the solvophobic character of FDTS layer mitigates side reactions between the anode and electrolyte, and enhances the desolvation rate at the interface. To confirm the stable presence of FDTS during cycling, X-ray photoelectron spectroscopy (XPS) analysis was performed on the

FDTS-Li anode after 50 cycles, as shown in Fig. 1d-g. The Si 2p spectrum reveals the presence of C-Si-Cl bonds (Fig. 1d) and the F 1s spectrum mainly displays C-F bonds (Fig. 1e), which are consistent with the FDTS molecular structure. These XPS results verify that the FDTS coating is successfully constructed onto the lithium metal anode and maintains stability after cycling process. After 20 nm sputtering, the F 1s spectrum (Fig. 1f) shows a considerable amount of C-F bonds and increased Li-F bonds, indicating the stable FDTS coating and the formation of LiF-rich inner SEI layer on the surface of lithium metal during the cycle (Fig. 1g). The SEI rich in LiF can prevent electron tunneling, reduce side reactions, inhibit lithium dendrite growth, and maintain the integrity of the SEI in the cycle. Raman spectroscopy was used to compare the FDTS-Li surface with pure FDTS, showing similar spectral peak shapes between the two, thereby confirming the successful FDTS coating on lithium metal (Fig. 1h). Scanning electron microscopy (SEM) image of FDTS-Li, along with energy-dispersive spectroscopy (EDS), indicates a homogeneous distribution of C, F, Si, and Cl elements (Fig. 1i), which is consistent with the elements in FDTS. In contrast, the Bare-Li surface exhibited negligible Si and Cl elements (Figure S5). These EDS results also validate the effective preparation of FDTS coating on the surface of lithium metal. To further analyze the reactivity of lithium metal and FDTS, we conducted surface composition analysis after immersing lithium metal in FDTS for 24 h (Figure S6). The XPS results show that the surface of lithium metal contains Si-Cl components, but no substances such as LiCl. Fig. 1j illustrates the results of Li⁺ desolvation energy on the anode surfaces of Bare-Li and FDTS-Li, as determined using the

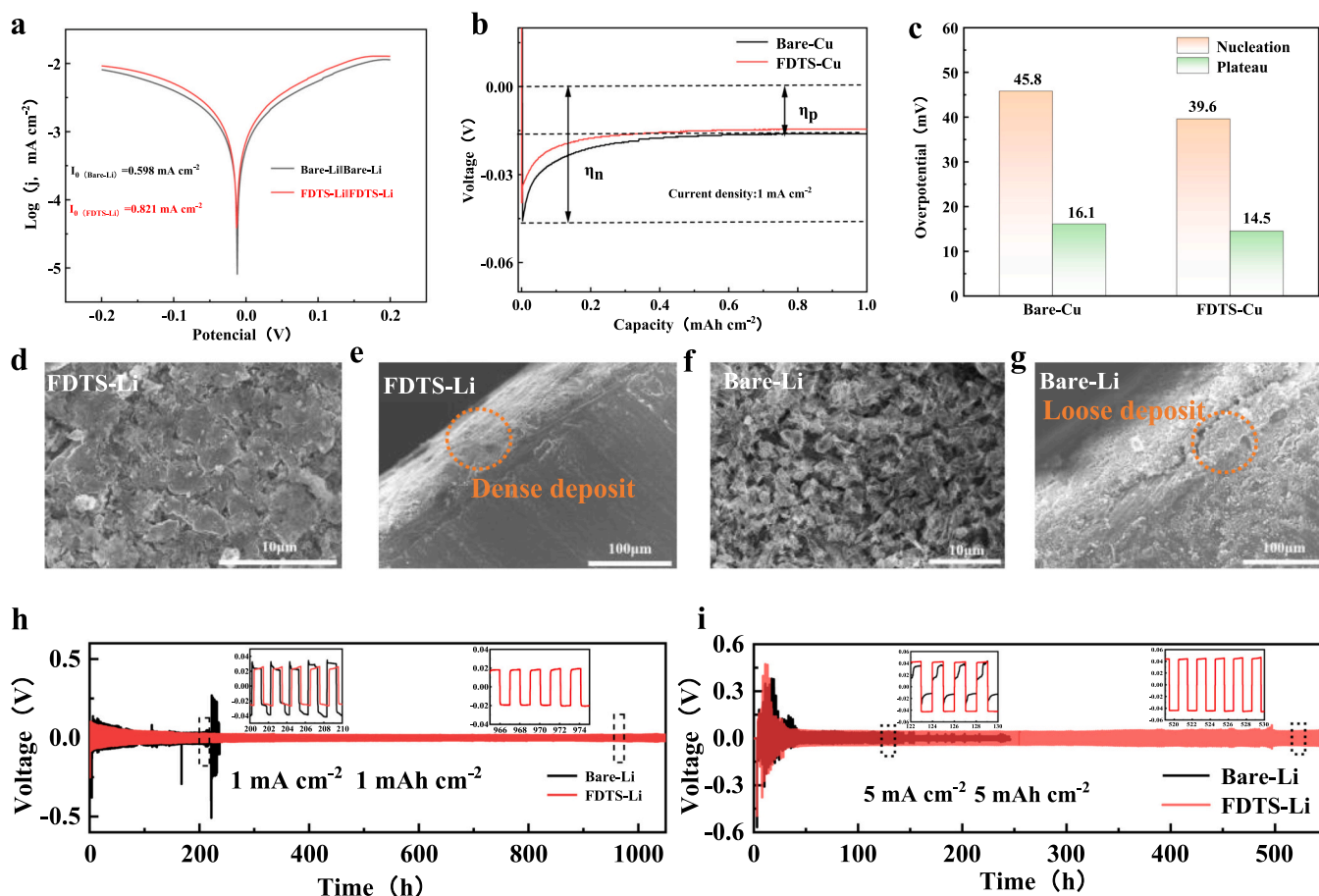


Fig. 2. Electrochemical performance characterization of Bare-Li and FDTS-Li anodes. a) Tafel curves for symmetrical Bare-Li and FDTS-Li cells and the corresponding exchange current densities of Bare-Li and FDTS-Li anodes. b) Galvanostatic voltage profiles and c) corresponding nucleation and plateau overpotentials of Li deposition on Bare-Cu and FDTS-Cu electrodes at a fixed current density of 1 mA cm^{-2} for 1 h. d-g) SEM images of surfaces and cross sections of Bare-Li||LFP and FDTS-Li||LFP batteries after 50 cycles at 3 C. Plating/stripping behaviors of symmetrical Bare-Li and FDTS-Li cells. h) Voltage-time profiles at a current density of 1 mA cm^{-2} and an areal capacity of 1 mAh cm^{-2} . i) Voltage-time profiles at a current density of 5 mA cm^{-2} and an areal capacity of 5 mAh cm^{-2} .

Arrhenius formula [27]. The impedance test results at different temperatures are shown in Figure S7. The desolvation energies for Bare-Li and FDTS-Li surfaces are 53.2 kJ mol^{-1} and $48.72 \text{ kJ mol}^{-1}$, respectively. The decreased desolvation energy confirm the more favourable desolvation process at the interface, leading to sufficient Li^+ supply for lithium deposition and achieving uniform deposition morphology. Subsequently, we further tested the air stability of the FDTS-Li, and the results are shown in Figure S8. Optical images and SEM results show that FDTS-Li is more stable in air and has strong hydrophobicity.

The electrochemical reaction kinetics were investigated using symmetrical cells composed of Bare-Li and FDTS-Li. As shown in Fig. 2a, the Tafel curve results reveal that the exchange current density of FDTS-Li is 0.821 mA cm^{-2} , compared to only 0.598 mA cm^{-2} for Bare-Li. This indicates a faster Li^+ transfer kinetic process on the FDTS-Li surface, because a higher exchange current density facilitates the plating and stripping of Li [28]. The deposition behavior of lithium on FDTS-Cu and Bare-Cu surfaces was further examined using Li||Cu half cells. Fig. 2b and c illustrate that the nucleation overpotential (η_n) and growth overpotential (η_p) of lithium on Bare-Cu are 45.8 mV and 16.1 mV , respectively, while on FDTS-Cu, these values are reduced to 39.6 mV and 14.5 mV , respectively. Similarly, on Bare-Li, the η_n and η_p are 379 mV and 66 mV , respectively, whereas on FDTS-Li, they decrease to 251 mV and 54 mV , respectively (Figure S9). Lower nucleation and growth overpotentials are beneficial for the uniform Li^+ deposition [29]. Additionally, the ion migration number of lithium ion across different surface was assessed, revealing a value of 0.82 for FDTS-Li, significantly higher than the 0.44 observed for Bare-Li (Figure S10). A higher lithium ion migration number diminishes the concentration gradient during charge process, thereby suppressing the growth of lithium dendrites [30].

To further verify the accelerated kinetic process of the FDTS-Li surface, which facilitates uniform lithium deposition, the lithium anode was examined using SEM after 50 cycles in a Li||LFP full cell. As shown in Fig. 2d-g, the FDTS-Li lithium anode exhibits dense and uniform lithium deposition on its surface, with the cross-sectional SEM image revealing a consistent and thin deposition layer. In contrast, the Bare-Li anode displays elongated dendritic growth, and the SEM images of the interface reveal uneven and dead lithium distribution. This difference is attributed to the faster transfer of Li^+ on the surface of FDTS. The stability of lithium metal was evaluated using symmetrical cells. Fig. 2h shows the deposition and stripping behavior of Bare-Li||Bare-Li and FDTS-Li||FDTS-Li symmetric cells at a current density of 1 mA cm^{-2} and a deposition capacity of 1 mAh cm^{-2} . FDTS-Li||FDTS-Li demonstrates stable operation for 800 h, maintaining a polarization voltage of approximately 18 mV . In contrast, Bare-Li||Bare-Li exhibits an increase in polarization voltage at 220 h, followed by a short circuit, which is attributed to uneven and irregular lithium deposition on the unmodified Bare-Li surface. This irregular deposition leads to the continuous formation of dendrites and dead lithium during long-term cycling, increasing polarization voltage and ultimately causing cell short-circuiting [31]. Fig. 2i illustrates the behaviour at a higher current density of 5 mA cm^{-2} and deposition capacity of 5 mAh cm^{-2} . Under these conditions, Bare-Li||Bare-Li short-circuits at approximately 140 h, due to the rapid growth of dendrites and penetrating the separator. Conversely, FDTS-Li||FDTS-Li operates stably for 550 h with a polarization voltage of about 40 mV . The improved cycle stability in symmetric cells also confirms the advantages of FDTS coating in promoting uniform lithium deposition. We conducted an average coulombic efficiency test and found that FDTS-Li has a coulombic efficiency of up to 99.55% , while the CE of Bare-Li is only 98.23% . A higher CE can achieve a more stable cycle. And we conducted a critical current density test and found that Bare-Li experienced a short circuit at a current density of 10 mA cm^{-2} , while FDTS Li was still able to function normally at a current density of up to 50 mA cm^{-2} , allowing the lithium metal negative electrode to remain stable under ultrafast charging (Figure S11).

To further investigate the impact of a solvophobic layer on lithium

surface deposition, we examined the deposition morphology under varying surface capacities and current densities. Fig. 3a and b illustrate the lithium deposition morphologies on Bare-Cu and FDTS-Cu surfaces, respectively, across different deposition capacities. As the deposition capacity increases from 0.1 mAh cm^{-2} to 100 mAh cm^{-2} , the FDTS-modified surfaces consistently exhibit dense lithium deposition. Additionally, we analyzed the initial lithium deposition morphology on surfaces with and without FDTS modification (Figure S12). On FDTS-modified surfaces, the lithium deposition is characterized by regular spherical shapes across a range of current densities, up to 6 mA cm^{-2} , whereas the Bare surface shows flocculent or even dendritic structures. The initial lithium deposition morphology on FDTS-modified surfaces aligns with previous reports under extreme conditions. Here, we attribute these observations to the following reasons: the solvophobic property of FDTS accelerates the desolvation of Li^+ at the interface, significantly reducing the solvent's influence on Li^+ deposition and increasing the Li^+ concentration at the interface. This facilitates the formation of an initial spherical deposition morphology when sufficient Li^+ is present and solvent influence is minimized. Consequently, the initial spheroid-forming nuclei evolve into denser lithium deposits rather than moss-like dendrites [32–34]. Meanwhile, to explore whether the initial spherical lithium deposition morphology is related to the electrolyte, we compared the lithium deposition morphology at different current densities in ether electrolyte and ester electrolyte (Figure S13). The results show that in ether electrolytes, due to the absence of the solvophobic effect at the interface, lithium deposition presents a dendrite appearance. Therefore, the initial spherical deposited metallic lithium obtained in this work is not related to the type of electrolyte, but is caused by the solvophobic effect of FDTS on the lithium metal anode.

To elucidate the impact of FDTS coating on lithium deposition, *in situ* optical microscopy (OM) was employed to examine the morphology of lithium deposits. The surface lithium deposition morphologies of Bare-Li and FDTS-Li, observed at a current density of 3 mA cm^{-2} over 0–20 min, are shown in Fig. 3c and d. On Bare-Li, moss-like dendrites emerged prominently within approximately 5 min, with subsequent dendrite growth becoming increasingly pronounced. Conversely, FDTS-Li consistently exhibits a dense and uniform deposition morphology devoid of dendrites. This difference is attributed to FDTS's ability to mitigate electrolyte side reactions and facilitate lateral lithium diffusion, resulting in dense deposits. The efficacy of FDTS in curbing dendrite growth and stabilizing the lithium metal interface is thereby underscored. Fig. 3e displays the surface conditions of Bare-Li and FDTS-Li post-cycling within diverse interface structures. The dendritic growth on Bare-Li is linked to the formation of an organic SEI, a consequence of the slow desolvation process. In contrast, FDTS-Li undergoes rapid desolvation, fostering a local environment abundant in anions and deficient in solvents, which leads to the generation of a rich inorganic SEI and dense lithium metal deposition.

The SEI composition is closely related to the Li^+ solvation structure at the interface [20,25,35], which can be precisely analyzed using Raman spectroscopy to determine the solvation state [36]. The Li^+ solvation structure is typically categorized into solvent-separated ion pairs (SSIP), contact ion pairs (CIP), and aggregates (AGG), which elucidate the coordination behavior of Li^+ . Fig. 4a presents the Raman spectra of the surface electrolytes on Bare-Li and FDTS-Li. The deconvolution analysis of these Raman spectra, as shown in Fig. 4b and 4c, indicates a significant increase in the CIP and AGG proportions on the FDTS-Li surface, accompanied by a decrease in SSIP structures. These findings indicate that a local solvation structure with high anion concentration and reduced solvent presence forms near the FDTS-Li surface, promoting the formation of an inorganic SEI. The SSIP content of Bare-Li is higher at the negative interface, which will lead to organic SEI derived mainly from solvent. The LiF-based inorganic SEI has high mechanical strength, inhibits dendrite growth, and enhances interface stability. Infrared spectroscopy further corroborates the transformation of the solution structure at the lithium metal interface after FDTS modification,

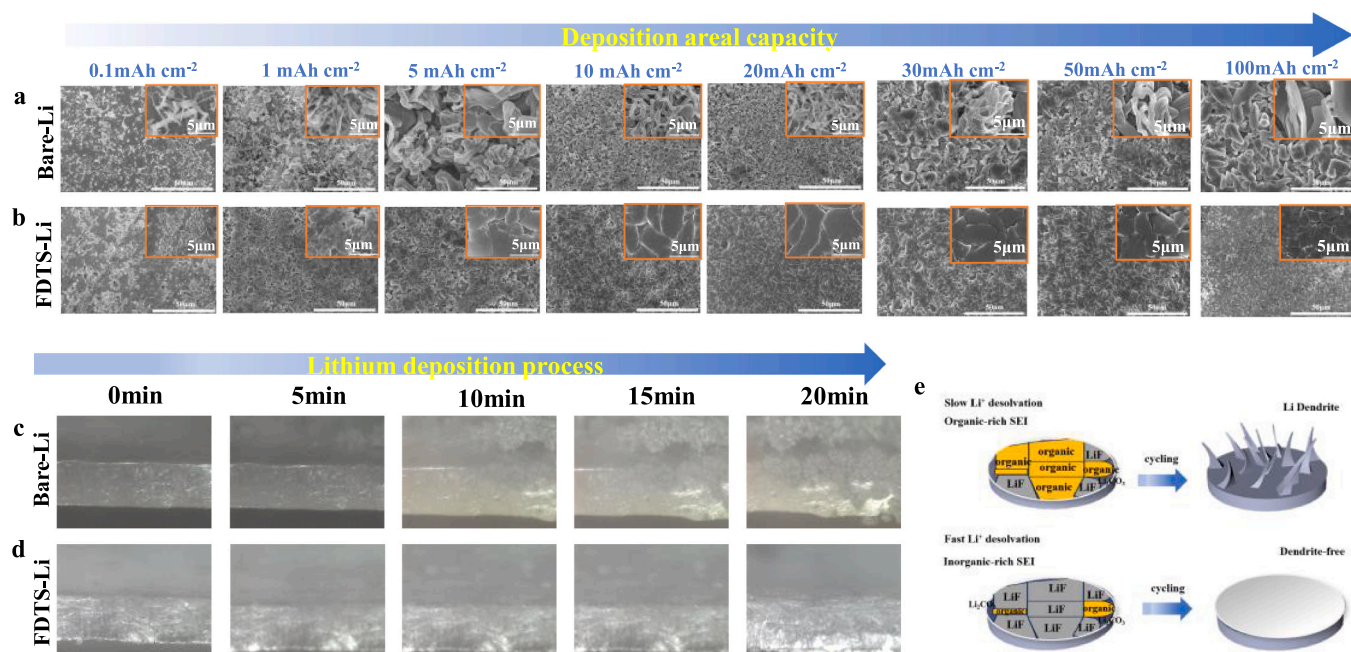


Fig. 3. Effect of FDTS modification on the morphology of Li deposition. SEM images of lithium deposition morphology under different deposition surface capacities at 1 mA cm^{-2} current density. a Bare and b FDTS. In situ OM images of Li deposition evolution on c Bare-Li and d FDTS-Li anodes after 0.5, 10, 15 and 20 min of discharge at a current density of 3 mA cm^{-2} . e Li surface state and schematic diagram of Bare-Li and FDTS-Li surface Li growth process after cycling.

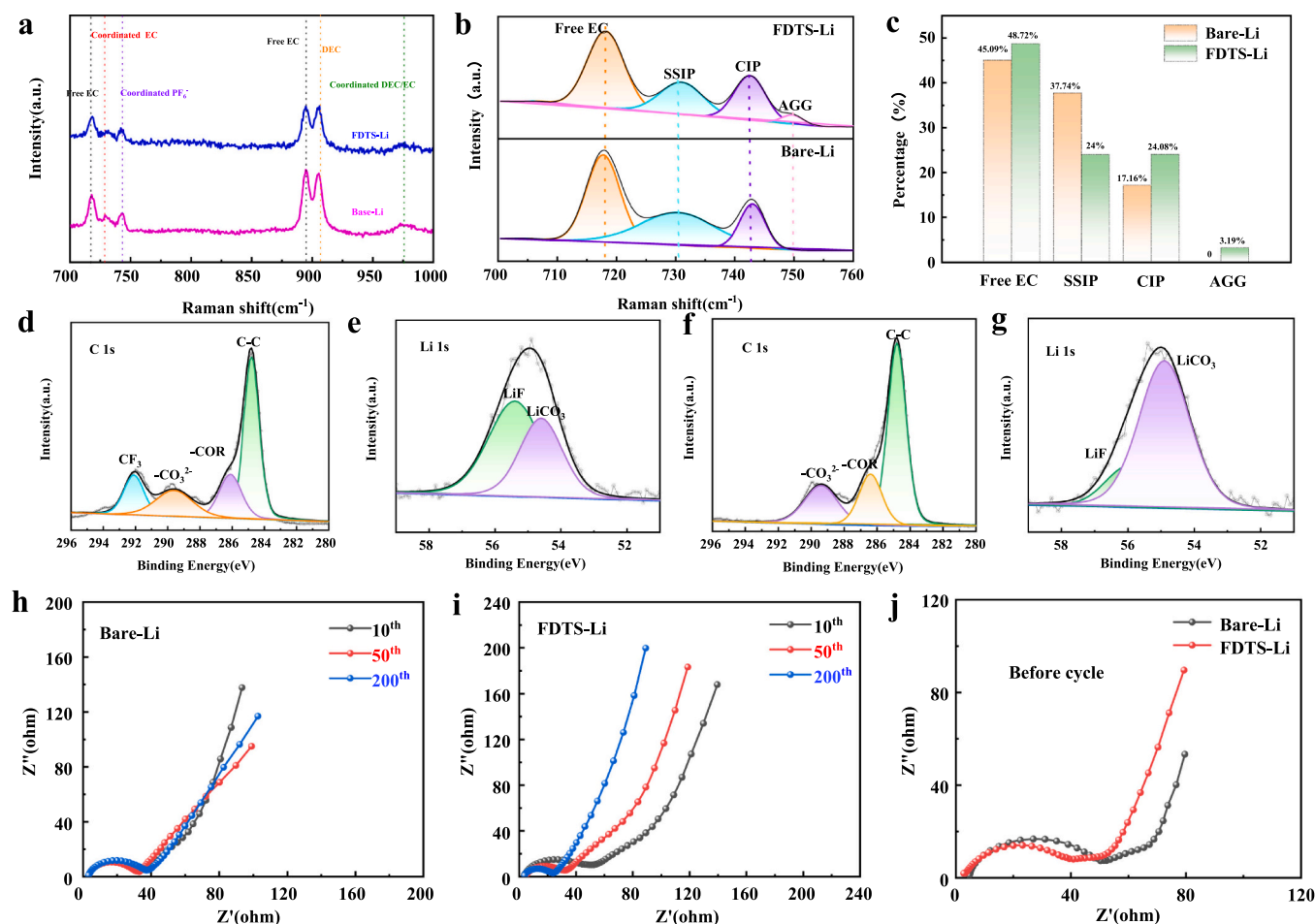


Fig. 4. a Raman spectra of Bare-Li and FDTS-Li. b Raman spectra of various solvation structures. c The percentage of different solvation structures. d C 1s and e Li 1s XPS spectra of FDTS-Li after 50 cycles. f C 1s and g Li 1s XPS spectra of Bare-Li after 50 cycles. EIS of the h Bare-Li||LFP and i FDTS-Li||LFP cells after 10, 50 and 200 cycles of the cell. j EIS of Bare-Li and FDTS-Li before the cycle.

revealing increased CIP and AGG structures, reduced solvent coordination, and enhanced anion coordination (Figure S14) [37]. XPS analysis, conducted after 50 cycles of Li||LFP, also detects changes in SEI components on the lithium anode surface. Fig. 4d and e display the C 1s and F 1s XPS spectra of the FDTS-Li surface, while Fig. 4f and g show the corresponding spectra for the Bare-Li surface. The results indicate that, compared to Bare-Li, the SEI on the FDTS-Li surface post-cycling is rich in LiF. The Cl 2p spectra of the XPS of the FDTS-Li cell after cycling is analyzed (Figure S15), and it is found that there is only Si-Cl on the anode surface and no LiCl component. Therefore, FDTS will not react with metallic lithium on its surface to form LiCl. FDTS does not decompose on the surface of the metal anode and participates in the formation of the SEI. Additional data in Figure S16 reveal the presence of Li_2CO_3 in the FDTS-Li surface SEI. Li_2CO_3 is contained in the SEI of both Bare-Li and FDTS-Li batteries, with no obvious difference. The XPS etching results further indicate that the SEI on the lithium metal surface of FDTS-Li cells has more inorganic components, while the SEI on the lithium metal surface of Bare-Li cells has more organic components (Figure S17). In summary, these findings indicate that the FDTS-modified lithium anode mainly improves the accelerated desolvation and transport of Li^+ at the interface and increases the LiF content through the interfacial kinetics of FDTS-Li during the cycling process, without changing the other components, thereby stabilizing the lithium metal anode.

The electrostatic potential energy (ESP) of solvent molecules such as EC, FEC, DEC, and FDTS was calculated, with the results shown in Figure S18. The unique molecular structure of FDTS results in a biased positive potential distribution across its entire molecular surface,

resulting in strong anion adsorption via electrostatic interactions and the formation of an anion-rich SEI. Electrochemical impedance spectroscopy (EIS) was employed to assess the charge transfer resistance of Bare-Li||LFP and FDTS-Li||LFP full cells after varying numbers of cycles, specifically after the 10th, 50th, and 200th cycles, as shown in Fig. 4h and i. The charge transfer resistance of Bare-Li||LFP increases by $10\ \Omega$ after 200 cycles compared to the 10th cycle. Conversely, the charge transfer resistance of FDTS-Li||LFP decreases to one-third of its value after 10 cycles. This reduction is attributed to FDTS enhancing the interfacial dynamic process, minimizing electrolyte by-products, and lowering interfacial impedance. On the Bare-Li surface, electrolyte by-products accumulate progressively during cycling, leading to increased impedance. Additionally, pre-cycling impedance tests, as shown in Fig. 4j, indicate that the FDTS-modified lithium metal surface exhibits lower interface impedance.

An in-depth analysis of the solvophobic layer mechanism was conducted. Molecular dynamics simulations were used to calculate the solvent distribution and Li^+ coordination components at the interface. As shown in Fig. 5a-d, the solvent distribution on the surface of Bare-Li is considerably more extensive compared to that on FDTS-Li. This observation is further corroborated by a cross-sectional snapshot in Figure S19, which illustrates a significant reduction in free solvent content on the surface of FDTS-Li due to the solvophobic layer. This reduction mitigates electrolyte corrosion and accelerates the interfacial desolvation process. Fig. 5e and f present the number density distribution functions for Bare-Li and FDTS-Li, respectively, further confirming the diminished solvent presence on the surface of FDTS-Li, indicating effective desolvation by the solvophobic layer. Additionally, Figure S20

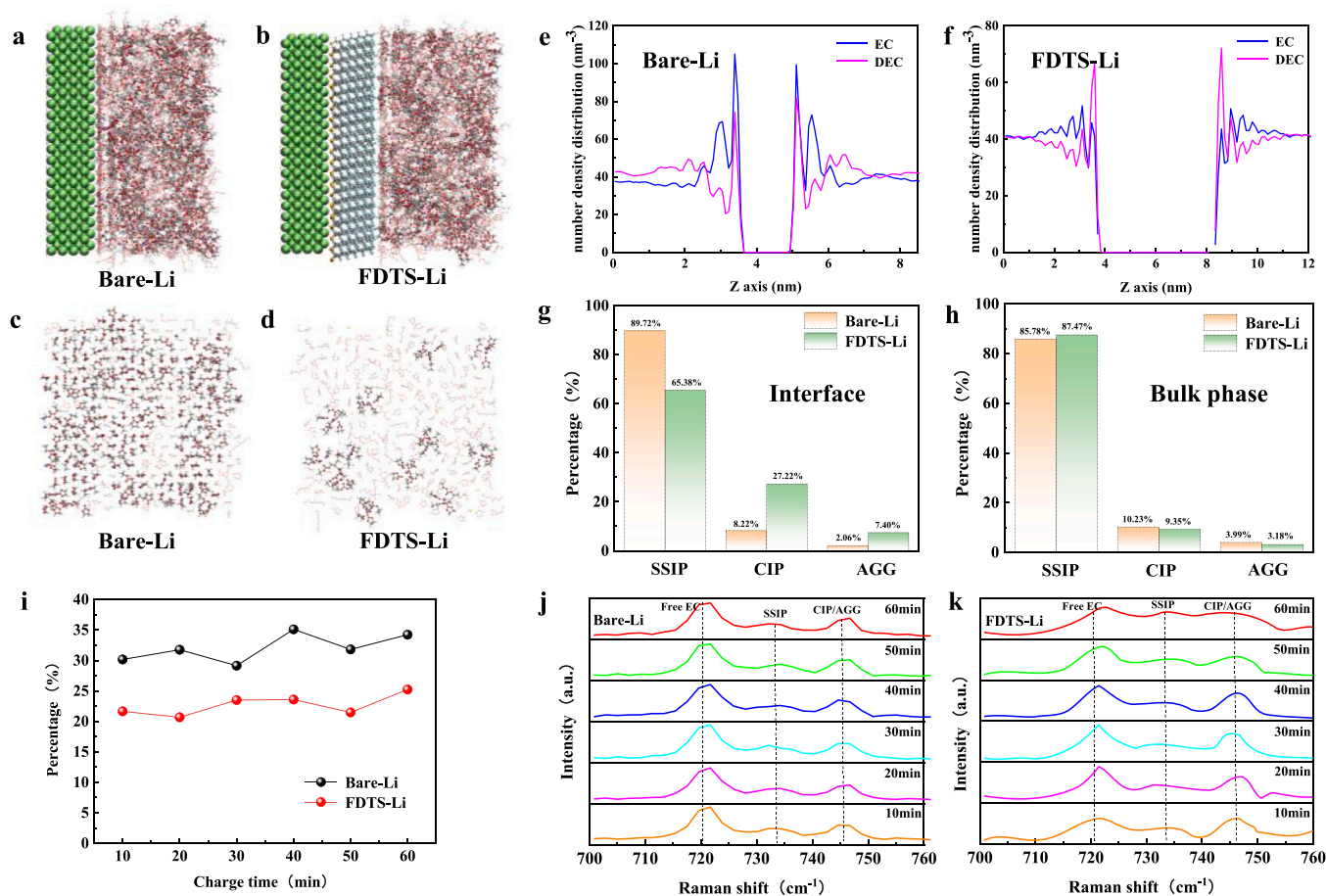


Fig. 5. Molecular dynamics simulation and in situ Raman spectroscopy. Solvent distribution at the interface of molecular dynamics simulation of a, c Bare-Li and b, d FDTS-Li. Number density distribution function of e Bare-Li and f FDTS-Li. g Solvation structure distribution at the interface. h Structure distribution diagram of solvation in bulk phase. i The ratio of Li^+ -solvent coordination structure at different charging times. In situ Raman spectroscopy tests on j Bare-Li and k FDTS-Li.

reveals an increase in Li^+ content on the surface of FDTS-Li, forming a sufficient Li^+ supply. The solvation structure distributions of the interface's near and far surface layers were also calculated, as shown in Fig. 5g and h. The proportion of CIP and AGG structures in the near surface layer of FDTS-Li increases significantly, whereas the solvation structure of the far surface layer remains largely unaffected for both Bare-Li and FDTS-Li. This indicates that the solvophobic layer primarily accelerates the desolvation kinetics at the interface and alters the solvation structure distribution there, without affecting the solvation structure of the bulk electrolyte phase, thereby maintaining high Li^+ conductivity in the bulk. The high content of SSIP in the bulk phase indicates that the bulk phase has a high ionic conductivity. Figures S21 and S22 provide radial distribution function (RDF) results for Li^+ at the interface and in the bulk, showing increased anion coordination at the FDTS-Li interface, with minimal differences in the bulk [38]. *In-situ* Raman spectroscopy, as referenced in Fig. 5i-k, characterizes the solvation structure ratio at the interface. The testing method and equipment for *in-situ* Raman spectroscopy are shown in Figure S23. The *in-situ* Raman test results indicate that the ratio of Li^+ -solvent coordination structure on the FDTS-Li surface remains consistently low during the charging process, suggesting rapid desolvation kinetics on the surface of FDTS-Li. Consequently, fast charging of lithium metal is achievable.

To evaluate the practical potential of the FDTS layer in protecting lithium anodes, a lithium metal full cell was constructed by pairing LFP with lithium anode. After modifying the lithium metal with FDTS, the lithium metal anode exhibited slightly larger interface polarization during the initial cycle. The EIS result after the first cycle is shown in Figure S24. At this stage, some Li^+ activated the interface to achieve rapid Li^+ transmission, showing a slight decrease in the initial capacity. However, after the subsequent stabilization, the capacity returned to the

normal level. Fig. 6a shows the long-term cycling performance of $\text{Li}||\text{LFP}$ full cells operating at 3 C. The full cell with FDTS-Li as the anode exhibits exceptional cycling stability, retaining 92.4% of its initial capacity after 2300 cycles. In contrast, the full cell with Bare-Li as the anode retains only 32.3% of its initial capacity after the same number of cycles. This indicates that the FDTS layer plays a critical role in ensuring the electrochemical cycling stability of the full cell. Figs. 6b and 6c present the corresponding charge-discharge curves for the FDTS-Li and Bare-Li full cells. The charge-discharge curves for the 1st and 2000th cycles of the FDTS-Li full cell remain nearly identical, whereas the Bare-Li full cell shows significant voltage and capacity decay. Further rapid charge-discharge tests at 10 C are shown in Fig. 6d, revealing that the full cell with FDTS-Li as the anode retains 97.7% of its initial capacity after 1500 cycles and maintains a Coulombic efficiency up to 99.9%. In contrast, the full cell with Bare-Li as the anode exhibits a short circuit after approximately 800 cycles, due to severe lithium dendrite growth and piercing the separator at high rates. This finding confirms that the FDTS layer regulates lithium deposition on the anode surface, reducing dendrite formation under fast-charge conditions. Figs. 6e and 6f display the corresponding charge-discharge curves for the FDTS-Li and Bare-Li full cells. The charge-discharge curves for the 1st and 1400th cycles of the FDTS-Li full cell are almost consistent, while the Bare-Li full cell exhibited significant overcharge and reduced Coulombic efficiency after 400 cycles. To further validate the role of FDTS during fast charging, ultra-fast charge-discharge tests at 20 C were conducted on $\text{Li}||\text{LFP}$ full cells, as shown in Fig. 6g. The full cell with FDTS-Li as the anode retain 85% of its initial capacity after 3000 cycles. While the full cell with Bare-Li as the anode displays a short circuit in a much shorter duration (around 400 cycles). The corresponding charge-discharge curves are included in Figure S25. This is attributed to the ultra-high charging rates, which leads to irregular lithium deposition and rapid dendrite

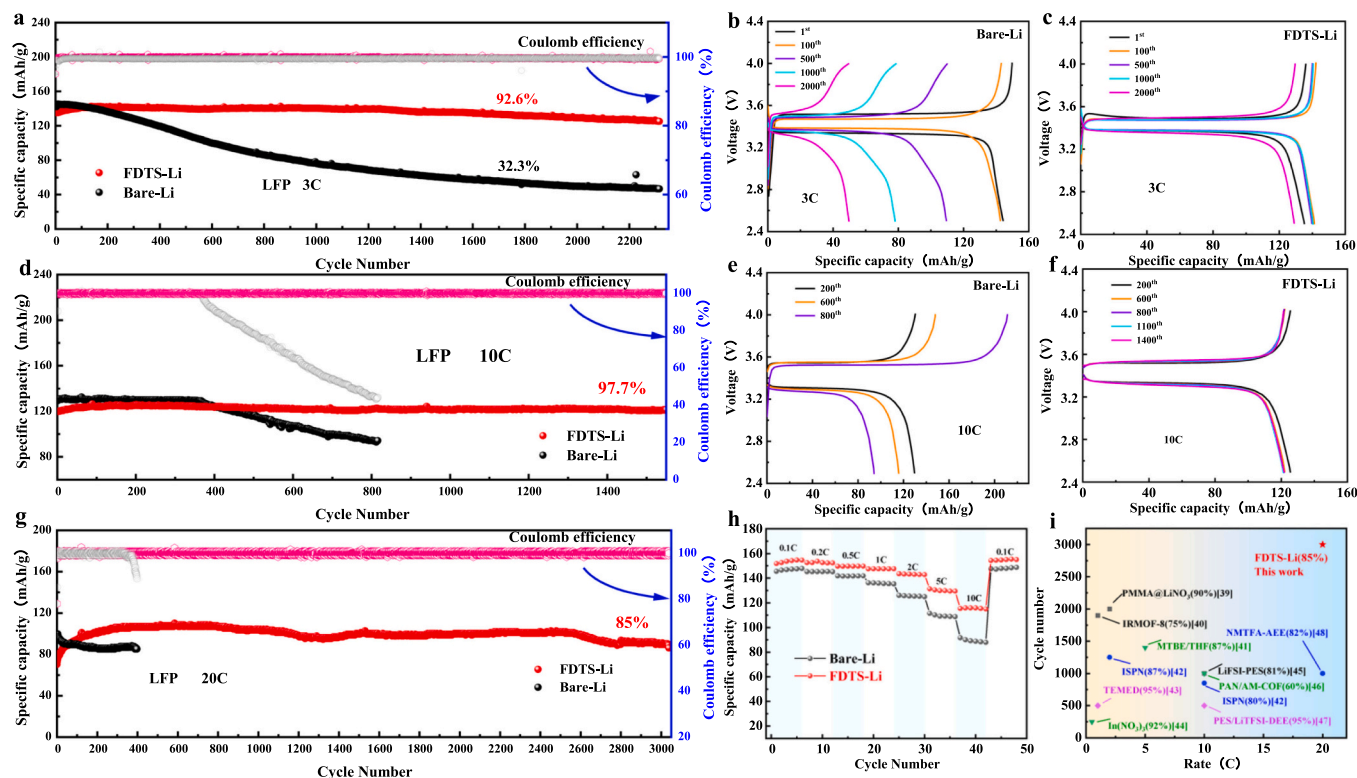


Fig. 6. Electrochemical performance of the $\text{Li}||\text{LFP}$ full cells. a Long-term cycling performance of the cells with Bare-Li and FDTS-Li at 3 C charge and 3 C discharge. Charge-discharge curves of the cell taken at the 1st, 100th, 500th, 1000th and 2000th cycles with b FDTS-Li and c Bare-Li. d Long-term cycling performance of the cells with Bare-Li and FDTS-Li at 10 C charge and 10 C discharge. Charge-discharge curves of the cell taken at the 200th, 600th, 800th, 1100th, and 1400th cycles with e FDTS-Li and f Bare-Li. g Long-term cycling performance of the cells with Bare-Li and FDTS-Li at 20 C charge and 20 C discharge. h Rate capability of cells with FDTS-Li and Bare-Li for cycling from 0.1 to 10 C. i Fast charge performance of $\text{Li}||\text{LFP}$ compared with recently published results [39–48].

growth that pierces the separator. And after modified with FDTs, the interface on lithium metal exhibits a high Li^+ concentration and uniform deposition, resulting in ultra-stable cycling life. Fig. 6h shows the rate performance test of $\text{Li}||\text{LFP}$ full cells from 0.1 C to 10 C, where FDTs-Li demonstrates superior rate performance. The charge-discharge curves at different rates are included in Figure S26. A comparison of this work with recent related studies on lithium metal under fast charging is presented in Fig. 6i, highlighting this work's advancement in ultra-high rate cycling performance. The detailed comparison of electrochemical performance is presented in Table S1. The comparison with the fluorosilane-based molecular modification layer is shown in Table S2. The FDTs-Li $||$ LFP cell retains 99.7% capacity after 500 cycles at 3 C with a medium loading (7.5 mg cm^{-2}) (Figure S27), with negligible voltage and capacity decay in the charge-discharge curves (Figure S28). Even at a high loading (10.5 mg cm^{-2}), the FDTs-Li $||$ LFP cell maintains 86.7% capacity retention after 150 cycles at 3 C (Figure S29). This work demonstrates remarkable long-term cycling stability at high rates. We further tested the full cell performance under thin lithium (50 μm), assembled LFP $||$ 50 μm -Li and NCM811 $||$ 50 μm -Li full cells respectively, and conducted long cycle tests. The results are shown in Figure S30 and Figure S31. LFP has almost no decay after 300 cycles. NCM811 $||$ 50 μm -FDTs-Li with high loading capacity (10 mg cm^{-2}) still has 85% capacity retention after 150 cycles, while NCM811 $||$ 50 μm -Bare-Li only has 73% capacity retention after 150 cycles. Overall, these full cell results demonstrate the significant potential of the FDTs layer as a protective solvophobic coating for lithium metal anodes in high-rate fast-charging LMBs.

3. Conclusion

In summary, a coupled regulation of interfacial desolvation behavior and LiF-rich SEI chemistry is realized on LMAs surfaces through the introduction of an FDTs layer, leading to improved interfacial stability in commonly used carbonate electrolytes, particularly under fast-charging conditions. The modified interfacial solvation environment facilitates Li^+ desolvation at the electrode surface and promotes enhanced anion participation near the lithium metal, favoring the formation of a compact, LiF-rich SEI that supports efficient interfacial ion transport. Benefiting from this synergistic interfacial regulation, a sufficient and continuous Li^+ supply can be maintained at the lithium metal surface to accommodate rapid lithium deposition, resulting in uniform and spherical lithium growth under high current densities. The spherical deposition morphology of lithium metal is maintained at the ultra-high current density of 6 mA cm^{-2} . And dense dendrite-free lithium growth can still be achieved at the ultra-high deposition surface capacity of 100 mAh cm^{-2} . The capacity retention rate of $\text{Li}||\text{LFP}$ full cell is 85% after 3000 cycles of 20 C ultra-high rate of charge and discharge. The SEI design with enhanced desolvation functionality in this work presents an effective strategy to improve the matching between Li^+ transport and lithium deposition reaction, enabling stable LMAs under fast charging.

CRediT authorship contribution statement

Yu Zhao: Conceptualization, Data curation, Formal analysis, Writing – original draft. **Xinyue Zhao:** Investigation. **Honghao Wang:** Investigation. **Meng Li:** Data curation. **Zeen Wu:** Data curation. **Zeping Liu:** Methodology. **Zhaoyu Chen:** Methodology. **Yu Gu:** Methodology. **Yu Zhang:** Conceptualization, Supervision, Methodology, Funding acquisition, Writing – review & editing. **Naiqing Zhang:** Methodology, Funding acquisition, Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2026.112073.

Data availability

Data will be made available on request.

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